Integrali impropri

1 Studiare la convergenza dei seguenti integrali impropri
(eventualmente in dipendenza dal parametro $a \in \mathbb{R}$)

1.1) $\displaystyle\int_0^{+\infty} \left(1 + \frac{1}{1+x^2}\right) dx$ **1.2)** $\displaystyle\int_0^{+\infty} \frac{2x}{1+x^2}\, dx$ **1.3)** $\displaystyle\int_0^{+\infty} \frac{1+x^7}{1+x^9}\, dx$

1.4) $\displaystyle\int_0^{+\infty} \frac{e^{-x}}{1+e^{-2x}}\, dx$ **1.5)** $\displaystyle\int_2^{+\infty} \frac{\sin x \log x}{x^{5/4}}\, dx$ **1.6)** $\displaystyle\int_0^1 \log x\, dx$ **1.7)** $\displaystyle\int_3^8 \frac{1}{\sqrt[6]{|x-8|^5}}\, dx$

1.8) $\displaystyle\int_0^1 \frac{1}{\log x}\, dx$ **1.9)** $\displaystyle\int_0^{\pi/2} \tan x\, dx$ **1.10)** $\displaystyle\int_1^{+\infty} \frac{\cos x \sin(1/x)}{\log^2 x}\, dx$ **1.11)** $\displaystyle\int_0^{+\infty} \frac{1}{2^{ax}}\, dx$

1.12) $\displaystyle\int_1^{+\infty} \frac{1}{\sqrt{x^4-1}}\, dx$ **1.13)** $\displaystyle\int_2^{+\infty} \frac{|x^2-8x+1|}{x^3-7} dx$ **1.14)** $\displaystyle\int_0^{+\infty} \{x\}\, dx$

1.15) $\displaystyle\int_{-\infty}^{-1} \frac{e^x}{x}\, dx$ **1.16)** $\displaystyle\int_0^1 \frac{1}{\sqrt[3]{x \sin x}}\, dx$ **1.17)** $\displaystyle\int_2^{+\infty} \frac{1}{x^a \log(x^3)}\, dx$

1.18) $\displaystyle\int_0^{+\infty} \frac{e^{ax}}{1+e^x}\, dx$ **1.19)** $\displaystyle\int_0^1 \frac{\log(1+\arctan^a x)}{e^x - \sin x - 1}\, dx$ **1.20)** $\displaystyle\int_0^1 \frac{\sin^2 x \, \log^a x}{1-\cos x}\, dx$

1.21) $\displaystyle\int_2^{+\infty} \frac{1}{x^a \log(x^3)}\, dx$ **1.22)** $\displaystyle\int_{-\infty}^{+\infty} \frac{x^5}{e^x + e^{-x}}\, dx$ **1.23)** $\displaystyle\int_0^1 \frac{\sin^2 x \, \log^a x}{1-\cos x}\, dx$

1.24) $\displaystyle\int_0^{+\infty} \frac{\sqrt[3]{\sin(\sqrt{x})}\sqrt{\arctan x}}{x^\alpha \sqrt{1+x+x^2}}$ **1.25)** $\displaystyle\int_2^{+\infty} \frac{1}{x^a \log(x^3)}\, dx$

2 Calcolare, in caso di convergenza, i seguenti integrali impropri

2.1) $\displaystyle\int_0^{+\infty} x e^{-x}\, dx$ **2.2)** $\displaystyle\int_0^{+\infty} 2^{-x}\, dx$ **2.3)** $\displaystyle\int_0^{+\infty} e^{-x} \sin x\, dx$ **2.4)** $\displaystyle\int_0^1 \frac{1}{1-x^2}\, dx$

2.5) $\displaystyle\int_{-\infty}^{+\infty} \frac{1}{1+x^2}\, dx$ **2.6)** $\displaystyle\int_0^{+\infty} \frac{x}{1+x^3}\, dx$ **2.7)** $\displaystyle\int_0^{+\infty} \frac{2x+5}{x^3+x^2-2}\, dx$

2.8) $\displaystyle\int_3^{+\infty} \frac{x}{x^3-8}\, dx$ **2.9)** $\displaystyle\int_2^{+\infty} \frac{1}{x \log^4 x}\, dx$ **2.10)** $\displaystyle\int_{-\infty}^1 \frac{x}{1+x^4}\, dx$ **2.11)** $\displaystyle\int_0^2 \frac{x}{4-x^2}\, dx$

2.12) $\displaystyle\int_0^1 \frac{1+\sqrt{x}}{\sqrt{x}}\, dx$ **2.13)** $\displaystyle\int_0^1 \frac{1}{1+\sqrt[3]{x}}\, dx$ **2.14)** $\displaystyle\int_0^1 \frac{\sqrt{1+x}}{\sqrt{x}}\, dx$ **2.15)** $\displaystyle\int_0^1 \frac{\sqrt{x}}{x^2-2x}\, dx$

2.16) $\displaystyle\int_0^{+\infty} \frac{3x}{\sqrt{(1+x^2)^5}}\, dx$ **2.17)** $\displaystyle\int_0^{+\infty} \frac{\arctan^3 x}{1+x^2}\, dx$ **2.18)** $\displaystyle\int_0^{+\infty} \frac{2}{\sqrt{x}\,(1+x)}\, dx$

2.19) $\displaystyle\int_0^{+\infty} \frac{\log(1+\sqrt{x})}{\sqrt{x}\,(2+\sqrt{x})^2}\, dx$ **2.20)** $\displaystyle\int_1^{+\infty} \frac{\log x}{(x-1)^{3/2}}\, dx$ **2.21)** $\displaystyle\int_0^{+\infty} \frac{\log(1+x^2)}{x^2}\, dx$

3 Determinare il dominio delle seguenti funzioni integrali e tracciarne un grafico qualitativo

3.1) $F(x) = \displaystyle\int_0^x e^{t^2}\,dt$ **3.2)** $F(x) = \displaystyle\int_1^x \frac{e^t}{t}\,dt$ **3.3)** $F(x) = \displaystyle\int_0^x e^{-t^3}\,dt$ **3.4)** $F(x) = \displaystyle\int_{-1}^x e^{-|t|}\,dt$

3.5) $F(x) = \displaystyle\int_0^x e^{-t^2}\log(1+|t|)\,dt$ **3.6)** $F(x) = \displaystyle\int_0^x \log|t|\,dt$ **3.7)** $F(x) = \displaystyle\int_0^x t^2\log|t|\,dt$

3.8) $F(x) = \displaystyle\int_0^x e^t\log t\,dt$ **3.9)** $F(x) = \displaystyle\int_1^x \frac{1}{t\sqrt{t-1}}\,dt$ **3.10)** $F(x) = \displaystyle\int_0^x \frac{1}{\sqrt[3]{t^2-t}}\,dt$

3.11) $F(x) = \displaystyle\int_2^x \frac{e^t}{\sqrt[3]{t-2}}\,dt$ **3.12)** $F(x) = \displaystyle\int_0^x \frac{1}{\sqrt{t^2-5t+6}}\,dt$ **3.13)** $F(x) = \displaystyle\int_0^x e^{\sqrt{t-t^2}}\,dt$

3.14) $F(x) = \displaystyle\int_0^x \frac{\log|t|}{1+t^2}\,dt$ **3.15)** $F(x) = \displaystyle\int_0^x \frac{1}{\sqrt[3]{1-t}\,(1+t^2)}\,dt$ **3.16)** $F(x) = \displaystyle\int_0^x \frac{1}{\sqrt{|t^2-1|}}\,dt$

3.17) $F(x) = \displaystyle\int_1^x \frac{e^{t-1}}{t^4}\,dt$ **3.18)** $F(x) = \displaystyle\int_0^x \frac{1+t^5}{\log t}\,dt$ **3.19)** $F(x) = \displaystyle\int_0^x \frac{\sin t}{1+t^2}\,dt$